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Determinantes Matrizes de Ordem 1, 2 ou 3

(Fotos nas páginas seguintes)

Exercícios 1 e 2:

D S T Q Q S S

Tarefa Básica - Determinantes.

1-) A-) $\begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$ $\begin{matrix} 3 \\ 10 \end{matrix}$ $\Rightarrow 10 - 3 = \boxed{7}$ m

B-) $\begin{bmatrix} -2 & -4 \\ 3 & 6 \end{bmatrix}$ $\begin{matrix} -12 \\ -12 \end{matrix}$ $\Rightarrow -12 + 12 = \boxed{0}$ m

C-) $\begin{bmatrix} 3 & -1 & 1 \\ 2 & 1 & -1 \\ 1 & 4 & -2 \end{bmatrix}$ $\begin{matrix} 3 & -1 \\ 2 & 1 \\ 1 & 4 \end{matrix}$

$1 + (-12) + 4 = \boxed{-7}$

$\Rightarrow 3 + 7 = \boxed{10}$ m

$-6 + 1 + 8 = \boxed{3}$

$\Rightarrow -3 + 3 + 16 = \boxed{16}$

D-) $\begin{bmatrix} 3 & 2 & -1 \\ 2 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}$ $\begin{matrix} 3 & 2 \\ 2 & 3 \\ 1 & 1 \end{matrix}$

$36 + 2 - 2 = \boxed{36}$

$36 + 16 = \boxed{20}$ m

2-) $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ $\begin{matrix} -3 & 0 \\ 0 & -3 \\ 0 & 0 \end{matrix}$ $\boxed{0}$ $\Rightarrow \boxed{-27}$

$\det A = -27 - 0 = \boxed{-27}$ m letra A.

Exercício 3:

D S T Q Q S S

$$\boxed{x^2+12x+9}$$

$$3-1 \quad \begin{vmatrix} x & 1 & x \\ 3 & x & 4 \\ 1 & 3 & 3 \end{vmatrix} = -3 \quad \begin{vmatrix} x & 1 & x \\ 3 & x & 4 \\ 1 & 3 & 3 \end{vmatrix} \begin{matrix} x & 1 \\ 3 & x \\ 1 & 3 \end{matrix} \Rightarrow 3x^2+4+9x-(x^2+12x+9)$$

$$\boxed{3x^2+4+9x}$$

$$3x^2+4+9x-(x^2+12x+9) \Rightarrow 3x^2+4+9x-x^2-12x-9=-3 \Rightarrow$$

$$\Rightarrow 2x^2-3x-5=-3 \Rightarrow \boxed{\underset{a}{2x^2}\underset{b}{-3x}\underset{c}{-2}}$$

$$\Delta=(-3)^2-4\cdot 2\cdot(-2)$$
$$\Delta=9+16$$
$$\Delta=25=\sqrt{\Delta}=5$$

$$\Rightarrow \frac{3\pm 5}{4} \Rightarrow \frac{8}{4}=\boxed{2}\ m$$

$$\Rightarrow \frac{-2}{4}=\boxed{-1/2}\ m$$

$$\boxed{\left\{-\frac{1}{2};2\right\}}$$ m letra E.

Exercício 4:

D S T Q Q S S

4-) $\begin{vmatrix} x-1 & -1 & 0 \\ 0 & x+1 & -1 \\ 2 & -1 & x+1 \end{vmatrix} = 2$

$\begin{vmatrix} x-1 & -1 & 0 \\ 0 & x+1 & -1 \\ 2 & -1 & x+1 \end{vmatrix}$ $\Rightarrow 2(x+1)\cdot 0 = \boxed{\emptyset}$

$\Rightarrow (x-1)\cdot(-1)\cdot(-1) \Rightarrow x+1\cdot(-1) = \boxed{-x-1}$

$\Rightarrow 0\cdot(-1)\cdot(x+1) = \boxed{\emptyset}$

$\begin{matrix} x-1 & -1 & 0 \\ 0 & x+1 & -1 \end{matrix}$

$\Rightarrow (x-1)\cdot(x+1)\cdot(x+1) \Rightarrow$

$\Rightarrow x^2+x-x-1\cdot(x+1) \Rightarrow$

$\Rightarrow \boxed{x^3+x^2-x-1}$

$\Rightarrow 0\cdot(-1)\cdot 0 = \boxed{\emptyset}$

$\Rightarrow 2\cdot(-1)\cdot(-1) = \boxed{2}$

PRIMÁRIA:

$x^3+x^2-x-1+0+2 \Rightarrow$

$\Rightarrow \boxed{x^3+x^2-x+1}$ m

SECUNDÁRIA $\Rightarrow 0+x-1+0 = \boxed{x-1}$ m

$x^3+x^2-x+1-(x-1)=2$

$x^3+x^2-x+1-x+1-2=0$

$x^3+x^2-2x=0$

a b c

$\frac{-b}{a} = x_1+x_2$

$\frac{-1}{1} = \boxed{-1} \Rightarrow \boxed{-1}$ m letra c.

Exercício 5:

5-) $A=(a_{ij})_{3\times2}$, $a_{ij}=2i-3j$.
$B=(b_{jk})_{2\times3}$, $b_{jk}=k-j$.

$$A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} 2\cdot1-3\cdot1 & 2\cdot1-3\cdot2 \\ 2\cdot2-3\cdot1 & 2\cdot2-3\cdot2 \\ 2\cdot3-3\cdot1 & 2\cdot3-3\cdot2 \end{bmatrix}$$

$$A=\begin{bmatrix} -1 & -4 \\ 1 & -2 \\ 3 & 0 \end{bmatrix}$$

$$B=\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$B=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$B=\begin{bmatrix} 1-1 & 2-1 & 3-1 \\ 1-2 & 2-2 & 3-2 \end{bmatrix}$$

$$B=\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$A\cdot B = A=\begin{bmatrix} -1 & -4 \\ 1 & -2 \\ 3 & 0 \end{bmatrix}$$

$$A\cdot B=\begin{bmatrix} 0+4 & -1+0 & -2-4 \\ 0+2 & 1+0 & 2-2 \\ 0+0 & 3+0 & 6+0 \end{bmatrix}$$

$$0+0-12=\boxed{-12}$$

$$A\cdot B=\begin{bmatrix} 4 & -1 & -6 \\ 2 & 1 & 0 \\ 0 & 3 & 6 \end{bmatrix}\begin{matrix} 4 & -1 \\ 2 & 1 \\ 0 & 3 \end{matrix} \Rightarrow -12+12=0$$

$$24+0-36=\boxed{-12}$$

$\boxed{\det A\cdot B=0}$ letra C

Exercício 6:

D S T Q Q S S

6-) $A = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}$

$A = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$ $B \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}$

$A \cdot B = \begin{bmatrix} 2+0+0 & -2+0-2 \\ -1-1+0 & 1+1+0 \end{bmatrix} = A \cdot B = \begin{bmatrix} 2 & -4 \\ -2 & 2 \end{bmatrix}$

$\det A \cdot B = \begin{bmatrix} 2 & -4 \\ -2 & 2 \end{bmatrix}$ 8 4 $\Rightarrow 4 - 8 = \boxed{-4}$ m

$\boxed{\det A \cdot B = \boxed{-4}\ m}$ m letra D.